UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

# Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Classes 003–004 · Total derivative, chain rule and implicit functions · compiled 14 October 2026

| | |
|---|---|
| ESA ★★★ | Asked repeatedly in past ESA papers |
| ESA ★★ | Asked in past ESA papers or computed in class-work problems |
| ESA ★ | Conceptual: taught in class, not examined so far |
| ADDED FOR CLARITY | Standard context that is not in the class material |
| TRANSCRIPT ONLY | Said in the lecture but not on any slide |
| VIDEO ONLY | Seen only in a frame of the class video |
| NOT ON CURRENT SLIDES | From older material or past papers, not this year's slides |
| S003·7 · S004·9 | Deck 003 / 004, slide number (rendered .pptx) |
| T003a 00:06:27 | Transcript 003 part I: “Total derivative, chain rule and implicit functions” |
| T003b 00:01:10 | Transcript 003 part II: “Problems on total derivatives, chain rule” |
| T004a 00:13:09 | Transcript 004 part I: “Implicit functions and problems” |
| T004b 00:02:31 | Transcript 004 part II: “Problems on composite and implicit functions” |
| CW U1 P14 · ESA Jan-24 s3 1b | Unit 1 class-work problem 14 · past-paper question |

## Contents


1. Figure G2.1 · Tree diagram for the total derivative
2. Figure G2.2 · The total derivative as the slope along a path
3. Figure G3.1 · The differential approximates the true change
4. Figure G4.1 · A point moving on a saddle and a cylinder
5. Figure G4.2 · Triangle with two sides and the included angle
6. Figure G5.1 · Tree diagram for the chain rule
7. Figure G6.1 · Branch diagram for implicit differentiation
8. Figure G6.2 · The curve $xe^y + ye^x = 0$ and its tangent at the origin
9. Figure G6.3 · Level curves and the slope $-f_x/f_y$


Colours follow the Okabe–Ito order: blue = main curve or first branch, vermilion = tangent, comparison or second branch, green = third series, purple = other directions. Dashed lines are tangents or approximations; grey dashed lines are references. Tree diagrams are redrawn from slides 003·5, 003·14 and 004·5. The 3-D picture of the point moving on the saddle and cylinder is drawn from the equations of slide 003·7.

# 2 The total derivative rule

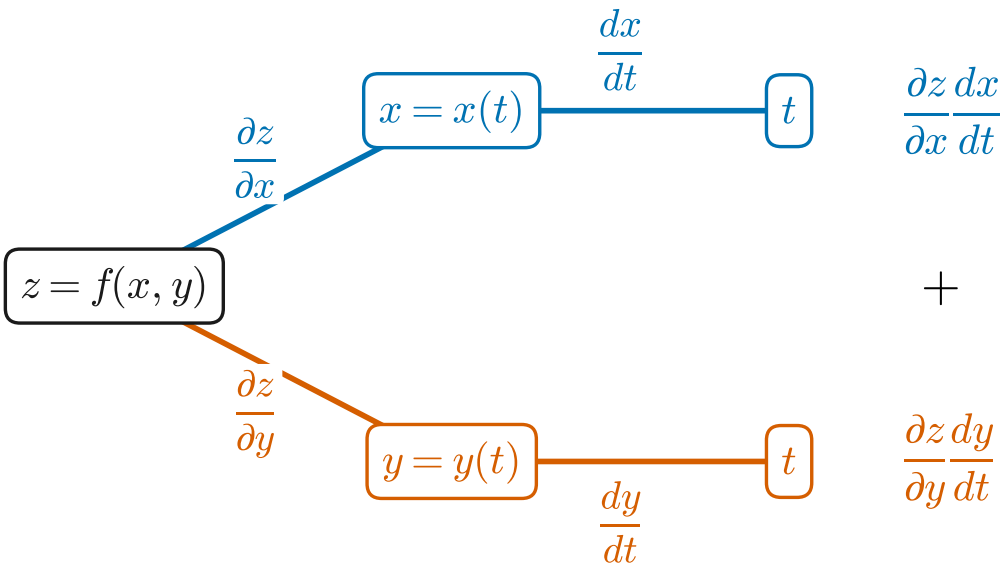



**Figure G2.1.** Tree diagram for the total derivative. ESA ★★

S003·5 · T003a 00:19:22

***How to read it.*** Start at $z$. Each branch carries the derivative of the upper variable with respect to the lower one. Multiply along each path from $z$ down to $t$, then add the two paths: $\dfrac{dz}{dt} = \dfrac{\partial z}{\partial x}\dfrac{dx}{dt} + \dfrac{\partial z}{\partial y}\dfrac{dy}{dt}$. In an exam, draw it first so that no path is missed.

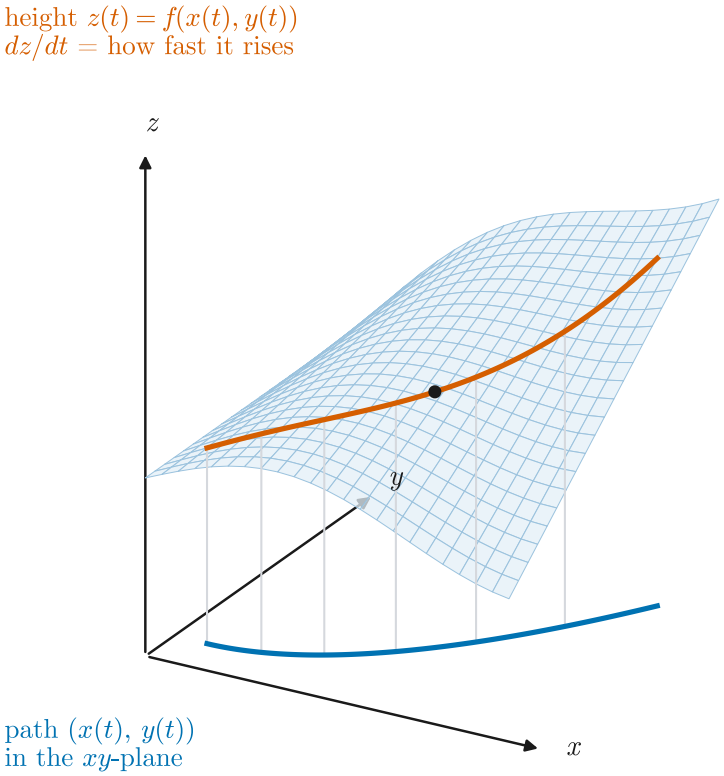



**Figure G2.2.** The total derivative as the slope along a path. ESA ★

ADDED FOR CLARITY S003·5 · T003a 00:05:20

***How to read it.*** As $t$ runs, the point $(x(t), y(t))$ moves along the blue path in the $xy$-plane. The vermilion curve is the surface height above it, $z(t) = f(x(t), y(t))$. $dz/dt$ is how fast that height changes as $t$ increases; it combines the slopes $z_x$ and $z_y$ weighted by how fast $x$ and $y$ move. The surface is an illustrative choice.

# 3 The total differential: approximation and sensitivity

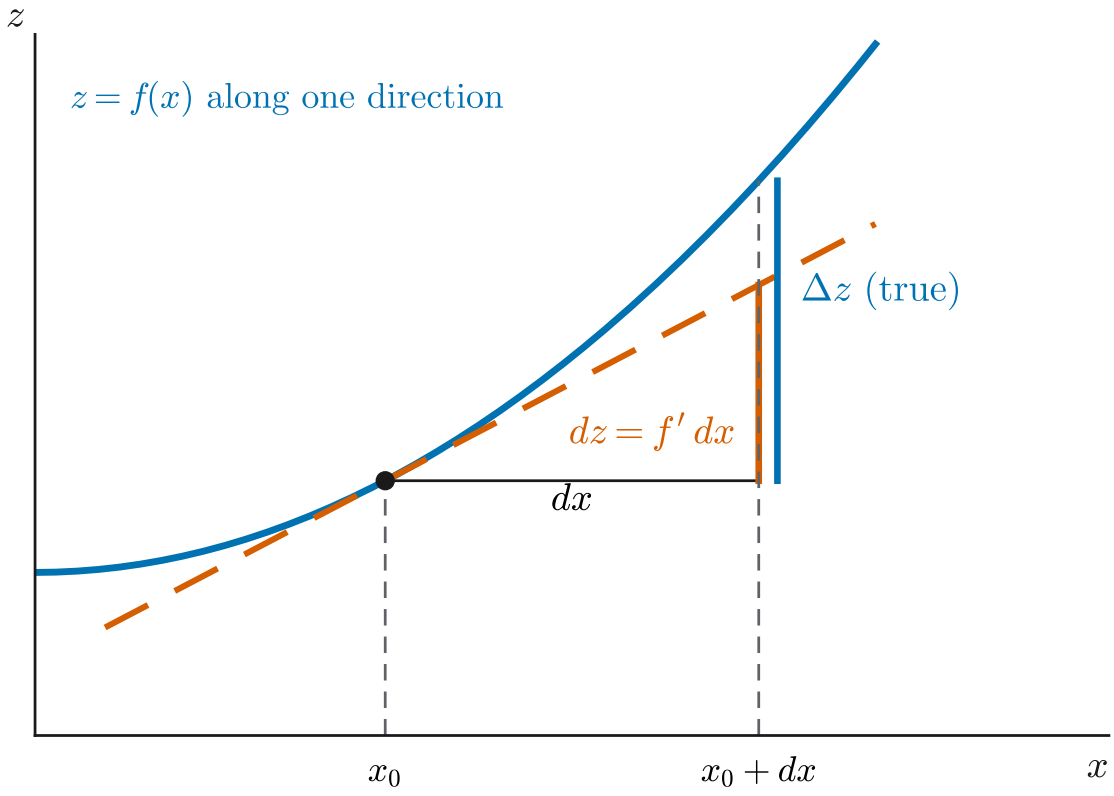



**Figure G3.1.** The differential approximates the true change. ESA ★★

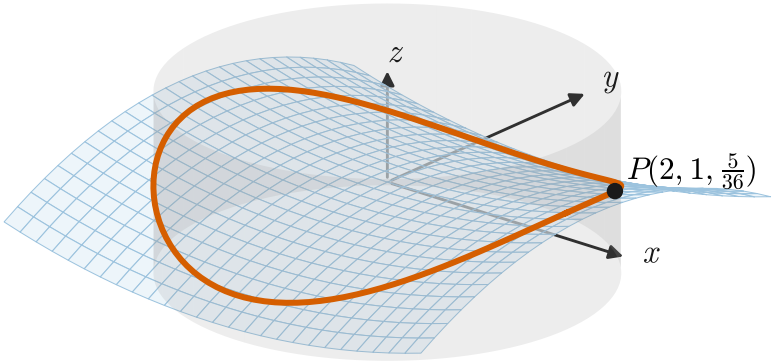
ADDED FOR CLARITY S003·8 · S004·4

***How to read it.*** Cut along one direction. The true change $\Delta z$ follows the blue curve; the differential $dz$ follows the dashed tangent line. For a small step $dx$ the two are nearly equal, which is why $f(x_0 + dx) \approx f(x_0) + dz$. The gap grows as the step grows.

# 4 Rates of change problems

saddle $z = \dfrac{x^2}{16} - \dfrac{y^2}{9}$

cylinder $x^2 + y^2 = 5$




intersection curve: path of $P$

**Figure G4.1.** A point moving on a saddle and a cylinder. ESA ★★

S003·7 · T003a 00:06:27

***How to read it.*** The saddle $z = x^2/16 - y^2/9$ (light blue) and the cylinder $x^2 + y^2 = 5$ (grey) meet in the vermilion curve. The point $P$ moves along that curve; at $x = 2$, $y = 1$ its height rises at $dz/dt = 5/36$ cm/s when $dx/dt = 0.2$ cm/s (W4.1). The slide calls the surface a paraboloid, but it is a hyperbolic paraboloid (a saddle).

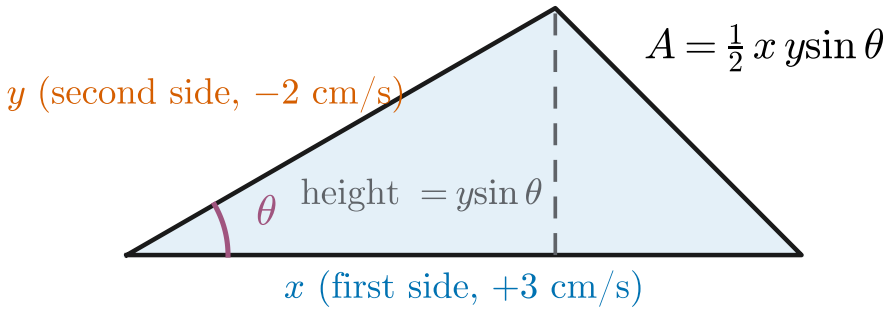



**Figure G4.2.** Triangle with two sides and the included angle. ESA ★★

S003·9 · T003a 00:15:25

***How to read it.*** Sides $x$ and $y$ meet at the angle $\theta$. The dashed perpendicular is the height $y \sin\theta$, so the area is $A = \frac{1}{2} x y \sin\theta$. Label the three changing quantities $x$, $y$, $\theta$ before differentiating.

# 5 The chain rule for composite functions

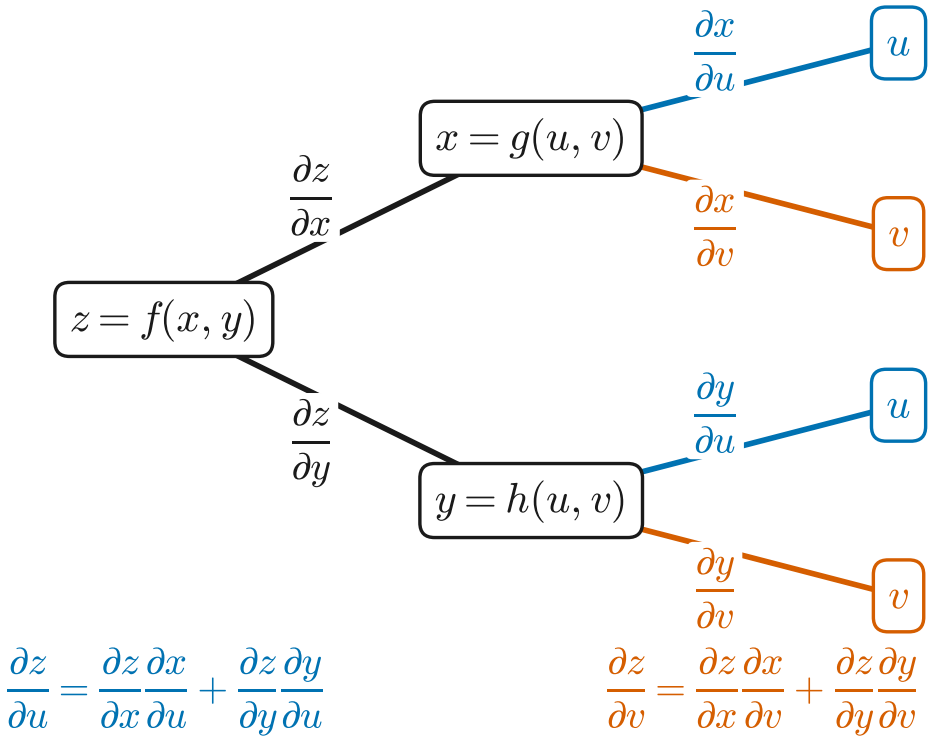



**Figure G5.1.** Tree diagram for the chain rule. ESA ★★★

S003·14 · T003a 00:19:42

***How to read it.*** $z$ branches to $x$ and $y$, and each of those branches to both $u$ and $v$, giving four leaves. For $\partial z/\partial u$ follow only the two paths ending at $u$ (blue): $z_x x_u + z_y y_u$. For $\partial z/\partial v$ follow the two ending at $v$ (vermilion). In an exam, draw the tree first and read the terms off it.

# 6 Implicit functions of two variables

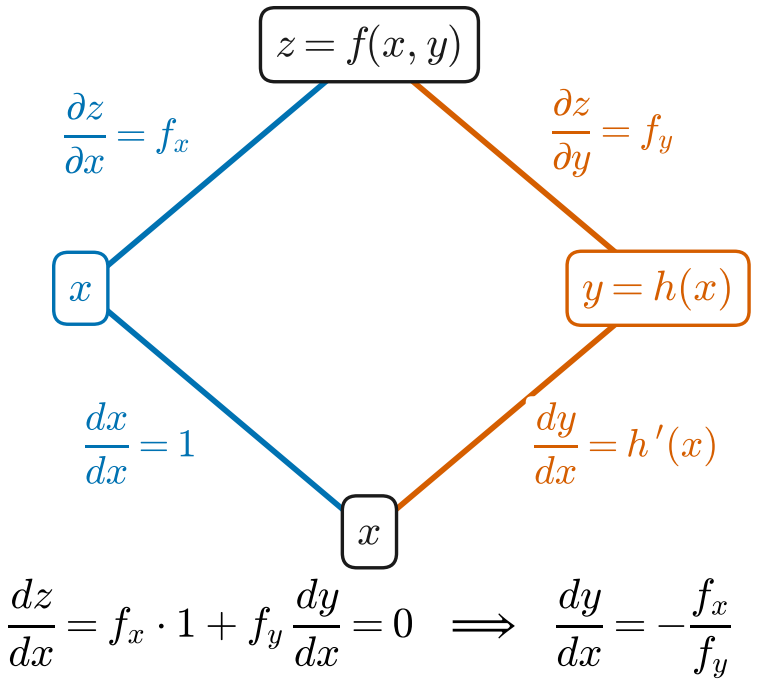



**Figure G6.1.** Branch diagram for implicit differentiation. ESA ★★

S004·5 · T004a 00:07:57

***How to read it.*** Differentiate $z = f(x, y)$ with respect to $x$ along both branches. The left branch is direct ($f_x \cdot dx/dx = f_x \cdot 1$). The right branch goes through $y = h(x)$ ($f_y \cdot dy/dx$). On a level curve the total is 0: $f_x + f_y\, dy/dx = 0$.

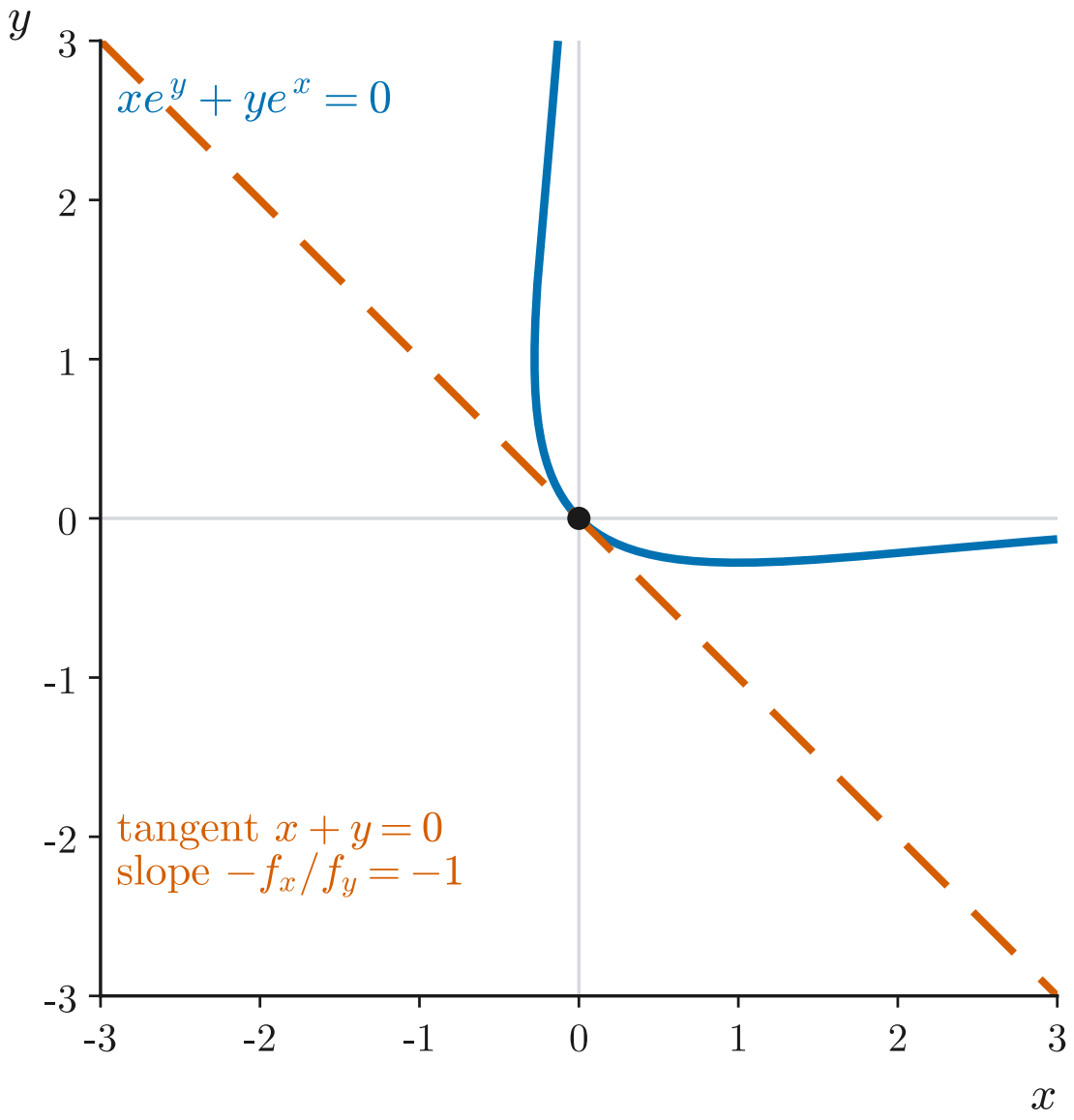



**Figure G6.2.** The curve $xe^y + ye^x = 0$ and its tangent at the origin. ESA ★★

S004·7 · T004a 00:13:48

***How to read it.*** The blue curve is the set where $f(x, y) = xe^y + ye^x = 0$. At the origin $dy/dx = -f_x/f_y = -1$, so the dashed tangent is $y = -x$. The curve is plotted from its equation (the slide's picture is a 3-D grapher screenshot).

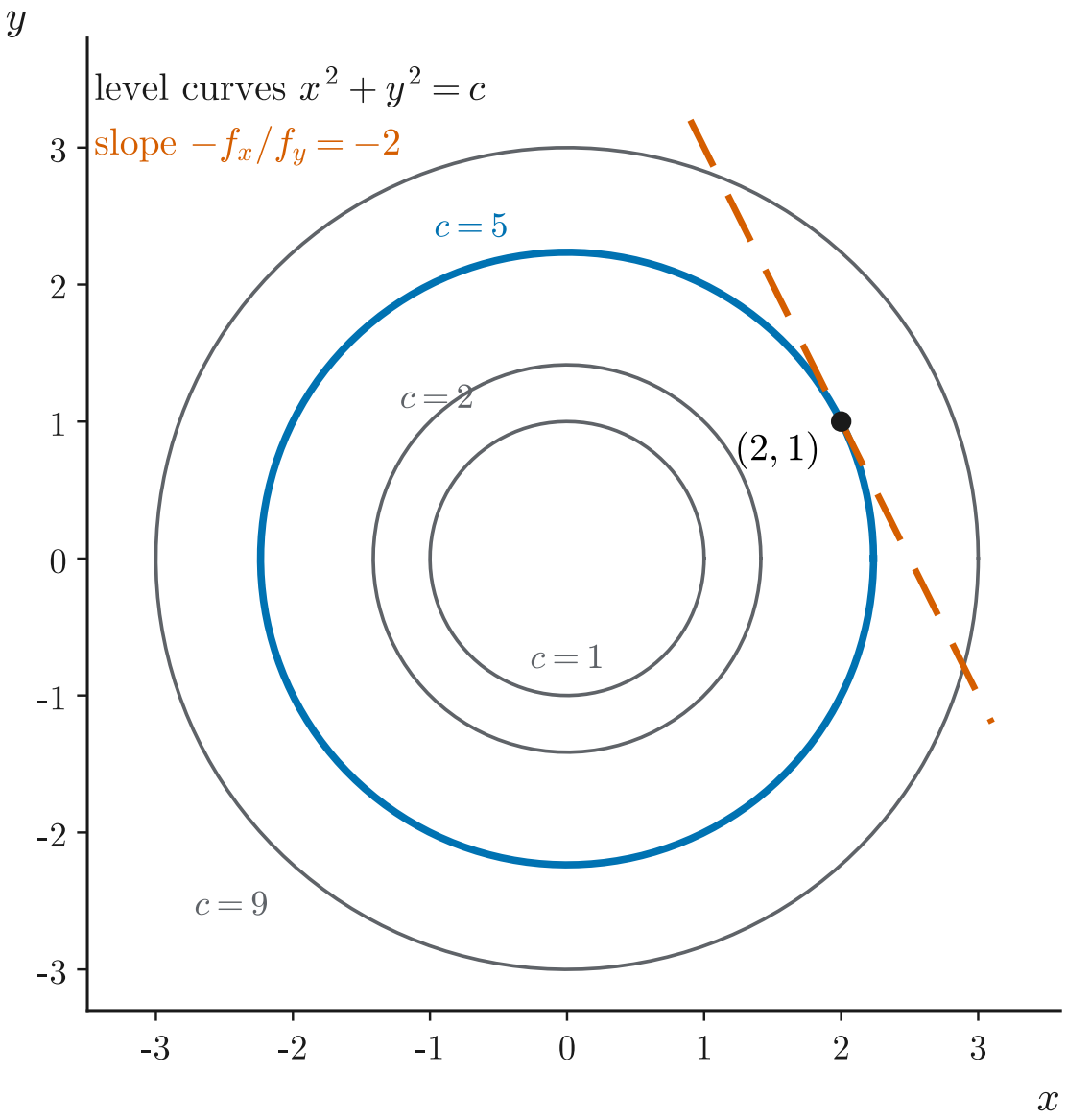



**Figure G6.3.** Level curves and the slope $-f_x/f_y$. ESA ★★

ADDED FOR CLARITY S004·4 · S003·7 · T004a 00:03:10

***How to read it.*** Level curves of $f = x^2 + y^2$ for $c = 1, 2, 5, 9$: the cross-sections of the cylinders on slide 004·4. On $c = 5$ at $(2, 1)$ the dashed tangent has slope $-f_x/f_y = -2x/2y = -2$, the same $dy/dx = -x/y$ used in W4.1.